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Soil contamination potential by trace metal elements in large-scale farms under different cropping systems in Western Kenya

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Abstract
Soil is one of the most important natural resources that play a central role in ensuring agricultural productivity. With the current increased application of agrochemicals, agricultural soils are under contamination threat in Kenya and other parts of the world. This study investigated the potential of trace metal elements contamination in agricultural soils under different cropping systems in large-scale farms in Trans Nzoia County of Western Kenya. From the selected farms, 204 soil samples collected were analyzed for pH, Electrical Conductivity (EC), exchangeable base cations (Na^+, Mg^{2+}, K^+, Ca^{2+}), Soil Organic Carbon (SOC), Total N, Cu, Pb, Zn, Cd, Fe and Cr. Base cations and trace metal elements were determined through Atomic Absorption Spectrophotometry while total N through calorimetry. Soil Organic Carbon was analyzed using Walkley-Black method. Statistical data analysis was done using Stata, version 15.0 while Trace Metal Element Pollution Potential (TMEPP) was computed by Metal Pollution Index (MPI). The study showed that the concentrations of Zn, Cu, Pb and Fe were lower than the globally acceptable levels in agricultural soils (MPI < 1) while Cd levels exceeded permitted levels (MPI > 1). Comparatively, maize and sugarcane cropping systems were the most susceptible to trace element pollution. In conclusion, the rank order of the pollution potential associated with the analyzed trace metal elements in the study area was Cd > Zn > Pb > Cu > Cr > Fe. The study recommends that there is need to regulate the application of Cadmium-containing agrochemicals and the types of crops cultivated in cadmium-rich soils to avoid possible transfer of this toxic metal to other trophic levels.

Keywords Cropping system, Contamination, Pollution index, Soil, Trace metal element

1 Introduction

Soil quality and characteristics play an important role in influencing agricultural productivity [1, 2]. Currently, soil pollution by trace metal deposition and accumulation in agricultural landscapes has become a global environmental concern [1, 3–10]. The effects of soil pollution do not only hamper the soil ecosystem's health but also extend to other associated ecosystems such as water, air, and human health through transfer of

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chemical residues by mechanical means as well as food chains and other trophic systems [2]. Need for enhanced agricultural production and emergence of various types of crop pests and diseases has necessitated increased use of mineral fertilizers and chemical pesticides in both developed and the developing world [11–13]. This is because while pesticides help in controlling crop pests and diseases [14–18], mineral fertilizers are believed to provide essential nutrients required by crops thus boosting yields [19–21]. Despite the high yields and harvest quality resulting from agrochemical application, the continued use of these inputs has been associated with a number of negative environmental externalities at various scales [2] including trace metal elements deposition [22].

From agronomic point of view, many of these trace metal elements such as cadmium (Cd), lead (Pb) and mercury (Hg) are considered non-essential for plant health and growth while others like cobalt (Co), copper (Cu), chromium (Cr), manganese (Mn) and zinc (Zn) are metabolically important to plants as micronutrients, though in low concentrations [7]. In humans, copper (Cu), iron (Fe) and Zinc (Zn) are biologically essential in development of skeletal structure, maintenance of colloidal systems and regulation acid-base equilibrium thus required in small quantities [7]. However, others like lead (Pb), silver (Ag) and Cadmium (Cd) have various harmful biological effects even at low concentrations. Within the soil, these elements are naturally present at background concentrations from natural sources such as volcanic eruptions, degradation of metal complexes, tectonic movements and pedogenesis [6, 23–24]. However, environmental concentration of trace metal residues have been associated with anthropogenic activities such as agriculture, industrialization, mining, transportation, fossil fuel combustion and waste disposal [1, 25–29]. This has made high pollution risks of trace metal elements in agricultural soils attract global attention [1].

In rural and non-industrial regions like Western Kenya, it is believed that increased usage of chemical pesticides and inorganic fertilizers remain the major source of soil trace metal pollution [30–31]. Most of these agrochemicals contain varying concentrations of Pb, Cd, Cu, Zn, and As, based on their sources [6, 28]. For example, it has been repeatedly observed that application of phosphate-based fertilizers results in elevated levels of Cd, Cr, Cu, As, Pb, Mn, Ni and Hg while biosolids and manures are also rich in Zn, Cu, Ni, Pb, Cd, Cr, As, and Hg [11, 22, 28, 32]. Within the soil, these trace metal residues are non-biodegradable, persistent and highly toxic to plants and other soil biota posing high threats to soil quality [4–6, 11, 28, 33]. This can be inferred from altered soil ecosystem processes such as anomalous nitrogen transformation, modified microbial loops, carbon mineralization and reduced soil fertility [7] which in the long run affects crop performance. Enhanced concentrations of trace metals in elemental form, or combined with other compounds are highly toxic and hazardous [34]. These concentrations may exhibit spatial variations in agricultural soils, based on soil physicochemical properties that influence the transformation of metal species through precipitation/dissolution, adsorption, sorption, methylation/demethylation, reduction/oxidation and formation of metal complexes [6, 35].

The trace metals in cationic forms or in complexes may leach into the groundwater or be transported by surface runoffs to surface waterbodies like rivers and streams. This can be accelerated by the high annual rainfall of up to 1,700 mm received in western Kenya, where this study was done [36]. Following their continued accumulation, certain food crops can absorb these trace metal residues from the contaminated soils, compromising

the quality and safety of food crops grown in contaminated soils thus posing substantial health risks to humans and other animals that depend on the food crops [7, 37–42]. Despite the adaptive enzymatic antioxidative defence mechanisms evolved by the plants to fight phytotoxic effects associated with trace metals [7], high accumulation of these elements in tissues is toxic to the plants. Their phytotoxic effects include oxidative damage, anomalous genetic change, abnormal reproductive responses, and signalling in biosynthetic pathways.

In Kenya, previous studies have reported occurrence of trace metal element residues in various environmental compartments [8, 38, 43–45] and horticultural crops harvested from contaminated sites. Western Kenya, particularly Trans Nzoia County is well known for large-scale maize production, with an annual production of at least 2.5 tonnes/ha. Besides maize, the region has large-scale monocultures of coffee and sugarcane. In the large-scale farms, there is frequent application of large volumes of various pesticide classes and mineral fertilizers [36]. In the present study, we hypothesised that intensive monocropping systems under agrochemical application exhibit higher TME enrichment in large-scale farms. The objective of the study was to analyse soil pollution potential by trace metal element deposition in six different cropping systems in large-scale farms in Western Kenya.

2 Materials and methods

2.1 Area of study

Trans-Nzoia County covers a total area of 2495.6 Km^2 lying between latitudes 00°52' and 10°18´ north of the equator and longitudes 034°38´ and 035°23´ east of the great Meridian, Western Kenya (Fig. 1). The County has a total human population of 818,757. Administratively, the County is divided into five sub-counties namely Kiminini, Saboti, Kwanza, Cheranganyi and Endebes. In relation to climate, it has a cool temperature with mean maximum temperatures ranging between 23.4 °C and 28.4°C (February) and mean minimum temperatures ranging between 11.0°C and 13.5°C (September). It receives an annual rainfall ranging between 1000 mm and 1700 mm with the highest being received in the Western parts of Endebess, Saboti and Kiminini Sub-Counties and North Western parts of Cherang'any Sub-county. Western Kenya regions have almost similar agroecological characteristics. Trans-Nzoia County was chosen as the study area to represent an area that is largest maize grain producer in Kenya and has conducive climatic conditions and land ownership that favor large – scale crop monoculture practice.

2.2 Sampling design

Sampling was conducted in large farms selected from the five sub-counties. A reconnaissance was conducted in the study area that involved a visit to the local authority's office in each ward to seek permission for engagement of farm owners to be involved in the survey. Six major cropping systems were purposively selected per sub-county including Maize farm (MF), Orchard farm (OF), Sugarcane farm (SF), Coffee farm (CF), Pasture lands (PL) and non-farmed control sites (NFS) without recent history of farming activities that acted as the control sites. Purposive sampling was applied in selecting the farms where soil samples were collected. In each farm, diagonal transects, 10 m apart were laid along which 1 m x 1 m were marked. From each quadrat, three replicate soil samples were taken from every site using a soil auger at a depth of 0–20 cm below the surface.

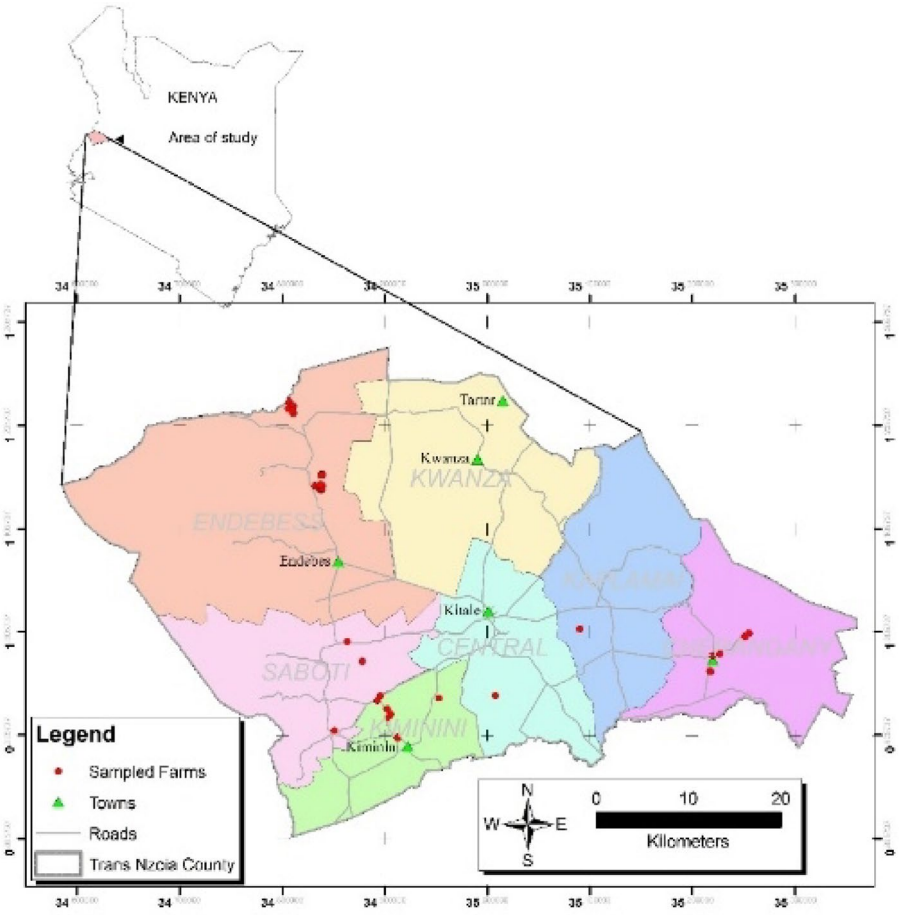

Fig. 1 Map of the study area

Similar quadrat sizes were marked outside the farms where there were no agricultural activities (NFS) that acted as control. The soil samples were labelled and transported to the laboratory for analysis at Jaramogi Oginga Odinga University of Science and Technology and University of Kabianga (Kenya).

2.3 Laboratory analysis of soil samples

2.3.1 Measurement of pH and electrical conductivity (EC)

Soil sample pH was measured on 2.5:1 soil to water suspension as described in [44]. To achieve this, 20 g of air dried soil sample was weighed and crushed using a mortar and pestle. The sample powder was sieved through a 2 mm sieve and transferred into a beaker. Fifty milliliters of deionised water was added to the beaker containing the sample and the mixture shaken for 10 min. The solution was allowed to stand for 30 min and stirred again for two minutes. pH values of the soil suspensions were measured using microprocessor pH meter, PHS-550, previously calibrated using pH 4 and pH 7 buffers. The EC of the extracts were read from the same suspensions after 24 h.

2.3.2 Determination of total nitrogen

Total nitrogen was determined by calorimetry method as presented in [44] that ensured no volatilization of any nitrogen. A selenium-based extraction mixture was prepared by dissolving 0.42 g selenium powder and 14 g lithium sulphate in 350 ml H_2O_2 in a 1000

ml beaker. The mixture was placed on ice in a water bath and 420 ml of concentrated sulphuric acid added gently. The solution was topped up to the mark with deionized water. Solution N1 was prepared by dissolving 34 g sodium salicylate, 25 g sodium citrate, and 25 g sodium tartrate in 750 ml distilled water. To this solution, 0.12 g of sodium nitroprusside was added and the solution made up to 1 L with deionized water. On the other hand, solution N2 was prepared by dissolving 30 g sodium hydroxide in 750 ml deionized water. After cooling, 10 ml sodium hypochlorite was added, mixed and the solution made up to 1 L.

Nitrogen/Littre standard solutions of concentration range 0, 2.5, 5.0, 7.5, 10.0 12.5 and 15.0 mg N/litre were prepared from 2500 mg N/litre stock solution. These standard solutions were treated as described in [44]. After the treatments, absorbance of the solution was measured using T170 UV/VIS spectrophotometer at 650 nm. From these absorbance, a standard calibration curve with a coefficient of determination (R^2) value of 0.99 were developed that was applied in quantifying the concentration of total N from the samples. Prepared soil samples were treated the same way as the standards. This involved measuring 0.3 g of finely ground soil sample into a digestion tube into which 4.4 ml of the prepared extraction mixture was added and digested for two hours at 360 °C until a colourless solution was obtained. The digest was allowed to cool and deionized water added until no more sediments could dissolve. The solution was transferred into a 50 ml conical flask and topped up to the mark with deionized water. After cooling, a clear solution was taken from the suspension. One millilitre of the sample digest was transferred into 9 ml of deionized water. Using a micropipette, 0.2 ml of sample digest and the blank were taken in a test tube into which 5 ml of reagent N1 was added and the mixture vortexed. After 15 min, 5 ml of reagent N2 was added and vortexed. The solution was allowed to stand for two hours after which the absorbance of the solution was measured using T170 UV/VIS spectrophotometer at 650 nm. The concentrations of total N in each of the samples was determined from the standard calibration curve.

2.3.3 Determination of soil organic carbon

Soil Organic Carbon was determined using the Walkley-Black method [47]. In brief, dry finely ground soil sample weighing 0.3 g was measured in a digestion tube into which 5 ml of 0.2 M potassium dichromate was added. To this solution, 7.5 ml concentrated sulphuric acid was added, and the mixture placed on a pre-heated block at 155 °C for 30 min. The content was allowed to cool and washed into a 200 ml conical flasks. To this solution, 0.3 ml prepared Ortho-Phenanthroline indicator was added. The solution was titrated, while stirring, against 2 M ferrous ammonium sulphate, until an end-point was reached with a colour change through greenish, blue to pink. Percent carbon was calculated using the formula described in [47].

2.3.4 Determination of base cations

Base cations (Na^+, Mg^{2+}, K^+, Ca^{2+}) were analyzed using modified methods by [44]. The samples were extracted with ammonium acetate solution to ensure maximum exchange between ammonium ions (NH_4^+) and the base cations. For every soil sample, 2.5 g of air dried soil was weighed into a clean stoppered plastic bottle. Fifty milliliters of 1 M NH_4OAc (pH 7) solution was added, content shaken using a mechanical shaker and filtered through No. 42 Whatman paper. This soil extract was used for the determination

of Na^+, Mg^{2+}, K^+, Ca^{2+} in $mg.kg^{-1}$. Na^+, K^+ and Ca^{2+} were determined using a dilution factor of 10, while Mg^{2+} was determined using a dilution factor of 25. Atomic absorption spectrophotometry, was undertaken using AAS type AI1200 in which the Method Detection Limit (MDL) was determined using Bernd Kraft certified reference material. The lamp current (mA) used for analysis of Na^+, K^+, Ca^{2+}, and Mg^{2+} were 12.0, 10 and 10, respectively with Air-C_2H_2 flame of detection limit ($mg.kg^{-1}$) of 0.006, 0.012, 0.07 and 0.003, respectively.

2.3.5 Determination of selected trace metal element concentrations

2.3.5.1 Quality control To ensure reliability of the results and avoid contamination during sample processing and instrumental readings, key quality assurance procedures were considered. For example, all the glassware used in the analyses were thoroughly washed in alkaline detergent, socked in nitric acid solution for 48 h and rinsed with deionized water. All the chemical reagents used in the analyses were of analytical grade obtained from Euro Chemical industries distributed by Twiga Chemicals and all solutions were made using deionized water as the solvent. In order to safeguard any effect of contamination during instrumental reading, two blank samples were run. The study analysed soil levels of Copper (Cu), Lead (Pb), Chromium III (Cr), Cadmium (Cd), Zinc (Zn) and Iron (Fe). These elements were determined according to the standard methods as outlined by [44] with slight modifications.

2.3.5.2 Sample preparation and instrumental analysis Five grams of air-dried soil sample was weighed and finely ground in a porcelain mortar using a pestle and sieved through 2 mm sieves. Two and a half grams of the powder was accurately weighed in to 100 ml beakers to which 20 ml of mixture of HCL and HNO_3 was added in the ratio of 3:1. The resulting mixture was digested on a hot plate and cooled. After cooling, the solution was diluted to 50 ml using deionized water. Concentrations of Copper (Cu), Lead (Pb), Chromium III (Cr), Cadmium (Cd), Zinc (Zn) and Iron (Fe) were determined from this solution according to the standard methods. Atomic absorption spectrophotometry was undertaken using AAS type AI1200 in which Method Detection Limit (MDL) was determined using Bernd Kraft® certified reference material. Cu, Pb, Cr, Cd, Zn and Fe were analysed using Air-C_2H_2 flame with detection limit ($mg.kg^{-1}$) of 0.04, 0.111, 0.089, 0.087, 0.008 and 0.076, respectively with calibration range ($mg.kg^{-1}$) of 0.1–0.6 and integration time of five seconds.

2.4 Data analysis

2.4.1 Inferential statistics

Statistical data analyses were done using Stata, Version 15.0. Skewness/Kurtosis tests for normality was undertaken that showed that the residuals depicted normal distribution. Descriptive statistics were computed on the concentrations of the trace metal elements, SOC, total N, base cations, pH and EC under various cropping systems. Test for normality of the data was determined using Shapiro-Wilk test. Multivariate analysis (MANOVA) was conducted to check significant differences among the trace metal element levels at various cropping systems. Spearman Rank Order Correlations were conducted to find out any significant correlations among the concentrations of the analyzed trace metal elements and other analyzed parameters.

2.4.2 Pollution index analysis

For each of the trace metal element analyzed, Metal Pollution Index (MPI) was also calculated according to a modified by [48];

$$\mathbf{MPI} = \frac{\boldsymbol{Cn}}{\boldsymbol{Sn}}$$

Where;

C_n; The concentration of the analyzed trace metal element in the sample;

S_n; The concentration of the trace metal element in the soil from control site.

Sites with MPI > 1 indicated surplus amount of the analyzed trace metal, thus high pollution potential while those with MPI < 1 indicated lower concentrations of the analyzed trace metal element thus less chances of pollution.

3 Results

3.1 Trace metal element concentration in various cropping systems

The various cropping systems considered in the study area recorded different levels of soil physicochemical parameters analyzed (Table 1). Skewness test showed normal distribution in the data (p-value of Skewness > 0.05). MANOVA showed significant differences in the means of the various analyzed parameters (F = 4.472;$p < 0.05$; Wilk's $\Lambda = 0.000$and partial $\eta^2 = 0.713$) as shown in Table 1, 2.

3.2 Variation in pH and electrical conductivity (EC)

The lowest mean pH (5.699 ± 0.390) was recorded under maize plantation while the highest was reported under coffee plantation. ANOVA showed a significant difference in the pH levels of soils sampled from the various cropping systems (F = 39.02; df = 203; $p \leq 0.05$).

The highest EC was recorded at the sugarcane plantation while the lowest was registered at the shrub land and sites where grass pastures were cultivated (Fig. 2). Differences in the means of EC at the various cropping systems were statistically significant (F = 36.05; df = 203; $p \leq 0.05$).

SOC also varied with cropping systems (Fig. 3). The highest mean of SOC (%)was observed at the pastureland and sugarcane plantations. The orchard and shrub land recorded the lowest TOC (%).

3.3 Variations in total N and base cations along cropping systems

The highest concentration of total nitrogen occurred at the pasture land while the lowest mean was reported from the soils sampled under sugar cane plantation and the shrub land (Fig. 4). Moreover, the mean concentrations of total nitrogen under the various cropping systems were significantly different (F = 36.678; df = 203; $p \leq 0.05$).

3.4 Metal pollution index (MPI)

The Metal Pollution Index (MPI) of the analyzed trace metal elements per cropping system was as shown in Table 3. With the exception of Cadmium, the concentrations of all the analyzed trace metals at each cropping systems were lower than the permitted global levels of these elements in agricultural soils (Table 3). In all cropping systems except coffee and pasture land, Cu and Zn concentrations tended towards pollution with

Table 1 Mean (± SD) of the concentrations of the analyzed soil chemical parameters in various cropping systems

Parameters	*N*	Maize	Orchard	Sugar cane	Coffee	Pasture land	Shrub land
pH	204	5.699 ± 0.390	6.636 ± 0.241	5.760 ± 0.279	6.491 ± 0.317	6.648 ± 0.179	6.640 ± 0.133
EC (µS cm^{-1})	204	17.467 ± 6.610	42.766 ± 15.210	10.533 ± 7.401	30.645 ± 18.170	9.700 ± 4.864	10.187 ± 7.600
Total N (Mg L^{-1})	204	0.415 ± 0. 362	0.769 ± 0.295	0.183 ± 0.024	0.188 ± 0. 087	0.730 ± 0.268	0.174 ± 0.189
Na (mg.kg^{-1})	204	549.748 ± 31.073	425.71 ± 107.038	450.260 ± 54.456	533.540 ± 91.057	583.798 ± 24.515	433.290 ± 96.874
K (mg.kg^{-1})	204	359.059 ± 74.505	246.18 ± 23.729	301.594 ± 54.456	320.051 ± 42.2458	271.069 ± 15.411	234.941 ± 82.533
Ca (mg.kg^{-1})	204	683.094 ± 29.657	372.73 ± 43.053	439.669 ± 99.284	526.749 ± 90.814	363.954 ± 22.951	389.770 ± 166.840
Mg (mg.kg^{-1})	204	353.145 ± 22.778	255.39 ± 44.621	371.607 ± 67.731	401.313 ± 84.577	367.795 ± 367.795	259.621 ± 47.814
Cu (mg.kg^{-1})	204	33.985 ± 1.731	20.152 ± 5.0748	30.503 ± 11.531	17.855 ± 3.344	14.587 ± 2.647	20.010 ± 4.414
Pb (mg.kg^{-1})	204	39.389 ± 3.455	29.077 ± 5.6342	36.634 ± 8.528	31.544 ± 7.604	24.532 ± 3.902	30.105 ± 7.298
Cr III (mg.kg^{-1})	204	11.251 ± 0. 604	8.311 ± 1.155	11.656 ± 1.542	11.095 ± 2.661	8.255 ± 1.005	8.494 ± 2.010
Cd (mg.kg^{-1})	204	9.042 ± 0.582	4.475 ± 0.531	9.347 ± 1.827	8.644 ± 1.325	8.032 ± 1.064	4.535 ± 0.992
Zn (mg.kg^{-1})	204	49.478 ± 2.588	29.488 ± 1.880	41.134 ± 5.255	43.663 ± 6.306	27.543 ± 3.927	30.776 ± 9.471
Fe (mg.kg^{-1})	204	1583.833 ± 77.953	1365.9 ± 39.638	1474.104 ± 77.142	1519.611 ± 115.022	1201.513 ± 70.629	1377.254 ± 136.205
SOC (%)	204	5.089 ± 0. 596	4.233 ± 0.525	5.877 ± 1.262	4.970 ± 1.295	6.195 ± 0.687	4.287 ± 2.048

MPI values approaching 1 ($0.5 < MPI < 1$). Pb and Cr, however recorded lower MPI values, inferring low soil pollution potential ($MPI \leq 1$). Similarly, Fe concentration showed less chances of pollution with $MPI < 0.5$ except at the sugar cane plantations where the concentrations tended towards pollution ($0.5 < MPI < 1$). Of all the trace metals studied, Cadmium concentrations showed $MPI > 1$ in all cropping systems, indicating surplus levels, thus soil pollution. However, the highest MPI for Cd concentration was recorded under maize, sugar cane, coffee and pasture cropping systems while the lowest values were recorded at the shrubland.

3.5 Correlations between the analysed physicochemical parameters

Spearman Rank Order Correlation showed a number of positive and negative correlations among the analysed soil parameters (Table 4). All correlations were significant at $p < 0.05$. EC was significantly positively correlated with total N ($r = 0.256$; $p < 0.05$), Na ($r = 0.204$; $p < 0.05$), Cu ($r = 0.150$; $p < 0.05$), Pb ($r = 0.326$; $p < 0.05$), Cr ($r = 0.280$; $p < 0.05$), Cd ($r = 0.200$; $p < 0.05$), Zn ($r = 0.347$; $p < 0.05$), Fe ($r = 0.159$; $p < 0.05$) and SOC ($r = 0.160$; $p < 0.05$). Soil pH was negatively correlated with most of the parameters including EC (r =-0.510; $p < 0.05$), Cu (r =-0.542; $p < 0.05$), Pb (r =-0.414; $p < 0.05$), Cr (r =-0.316; $p < 0.05$), Cd (r =-0.284; $p < 0.05$) and Zn (r =-0.310; $p < 0.05$). The concentrations of the base cations analyzed showed positive correlations with one another, where there was significance ($p < 0.05$).

The concentrations of the trace metals were positively correlated with each other ($p < 0.05$) and the base cations except for Magnesium that showed significant negative correlation with Cu (r =-0.145; $p < 0.05$) and Pb (r =-0.178; $p < 0.05$). Total N showed significant negative correlation with K (r =-0.161; $p < 0.05$) and Mg (r =-0.191; $p < 0.05$).

4 Discussions

4.1 Variations in pH, EC and base cation distributions

pH, negative logarithm to base ten of the concentration of hydrogen ions in a solution, is one of the basic soil properties that influence agricultural production. For example, pH has been found to influence microbial activity in biodegradation of organic wastes and mobility of certain nutrients and pollutants such as trace metal elements within the soil sub system. While soil acidity can be attributed to many variables in the environment such as high acidic rainfall, use of inorganic fertilizers and oxidative weathering, excess use of inorganic fertilizer can be the major contributor to low soil pH in agricultural landscapes such as western Kenya where this study was conducted. Therefore, from this study, lower soil pH values recorded under maize and sugarcane cropping systems (Fig. 5) is attributable to high application of inorganic fertilizers. Due to continuous farming and soil exhaustion, many large-scale famers in the study have resorted to periodic application of inorganic fertilizers [36]. Excess fertilizer residues within the soil results in enhanced production of hydrogen ions. The higher the concentrations of hydrogen ions held through the exchange of complex soil processes in relation to base cation concentration, the more acidic the soil sample becomes. For example, some inorganic nitrogenous fertilizers contain ammonium ions (NH_4^+) which during nitrification into NO_3^- is accompanied by the production of hydrogen ions released into the soil.

The mineralization and transfer of the hydrogen ions are accelerated by the high rainfall experienced in the study area. Under maize cropping systems, there is annual land

Table 2 Normality test for the data

Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Multivariate Tests[a]							
Intercept	Pillai's Trace	0.999	16127.179[b]	14.000	119.000	0.000	0.999
	Wilks' Lambda	0.001	16127.179[b]	14.000	119.000	0.000	0.999
	Hotelling's Trace	1897.315	16127.179[b]	14.000	119.000	0.000	0.999
	Roy's Largest Root	1897.315	16127.179[b]	14.000	119.000	0.000	0.999
Plots	Pillai's Trace	8.644	3.043	980.000	1848.000	0.000	0.617
	Wilks' Lambda	0.000	4.472	980.000	1701.707	0.000	0.713
	Hotelling's Trace	62.997	7.530	980.000	1640.000	0.000	0.818
	Roy's Largest Root	21.350	40.260[c]	70.000	132.000	0.000	0.955

[a]Design: Intercept + Plots

[b]Exact statistic

[c]The statistic is an upper bound on F that yields a lower bound on the significance level

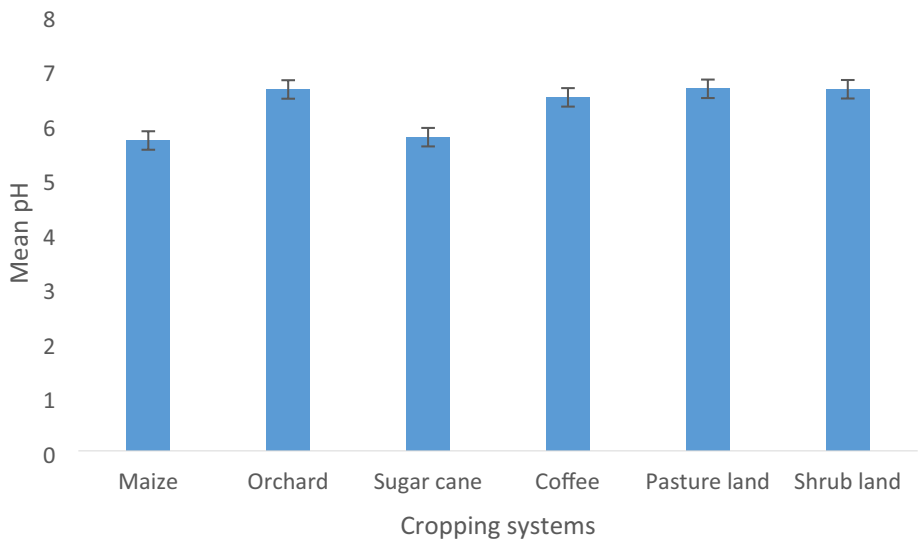

Fig. 2 Variation in Mean pH along cropping systems

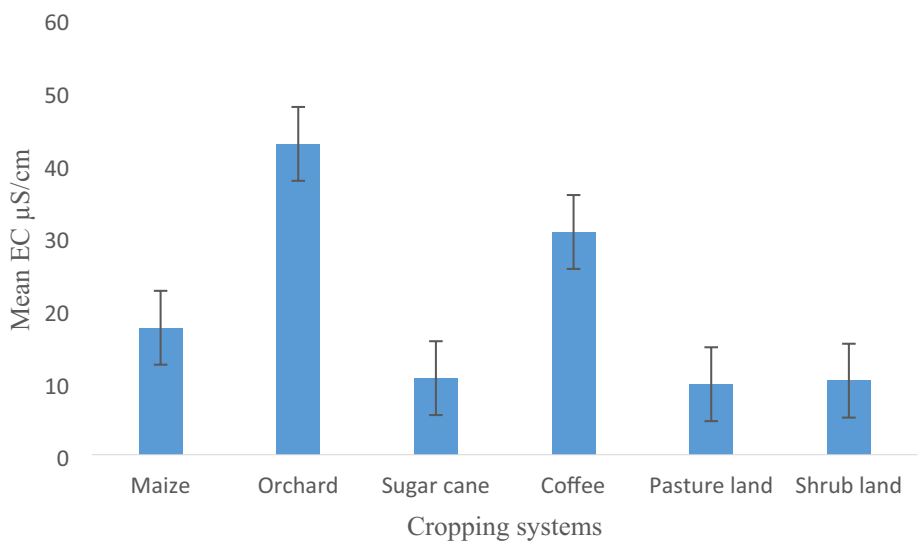

Fig. 3 Variation in mean EC across cropping systems

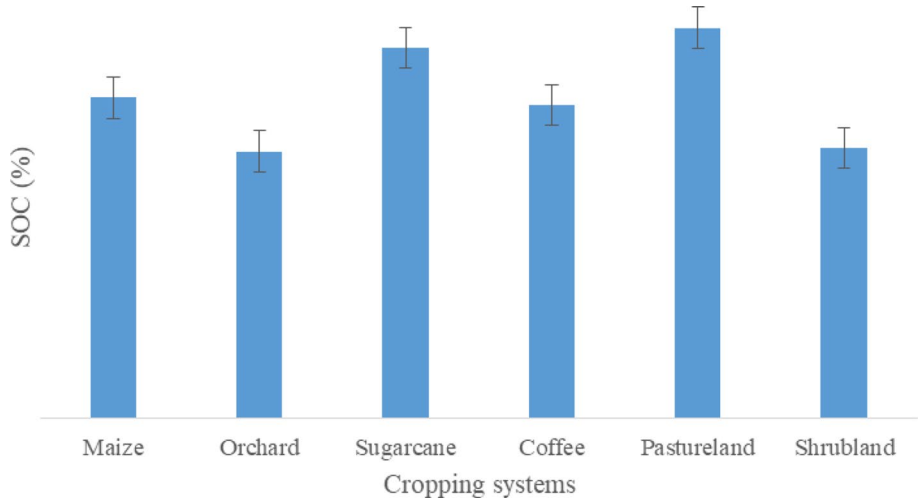

Fig. 4 Variation in mean SOC (%) across cropping systems

tillage which facilitates leaching of base cations to lower soil layers, leaving H^+ and $Al^{3+,}$ which are responsible for high acidity [49]. Under sugar cane cropping systems, there is regular application of inorganic fertilizers to increase yields. Fertilizer residues, coupled with high mineralization of N under cane canopies increase H^+ loads hence lowered pH. Contrastingly, there is infrequent application of inorganic fertilizers in the orchards and coffee farms. Low production of hydrogen ions thus leads to relatively higher soil pH compared to other cropping systems (Fig. 5). Moreover, it is important to note that Na^{2+}, K^+, $Mg^{2+,}$ and Ca^{2+} cations are basic. An increase in their concentration is therefore accompanied by a rise in pH. Ironically, the shrub lands in the study area that were considered pristine and less disturbed equally recorded lower pH implying acidity. Similarly, the pasture land with the paddocks recorded lower pH. The acidity of soil samples from the pasture land and shrub land areas can attributed to the presence of large deposits of organic matter/litter from plant leaves, decomposition and mineralization of which is accompanied with hydrogen ions production. This concurs with [50] that reported lower pH levels in grasslands. Additionally, the pristine shrub lands where sampling was done for control purposes were not fully vegetated. The exposure of soil to the heavy rainfall experienced in the study are could possibly facilitate heavy leaching of the exchangeable base cations leading to lower pH. Part of the hydrogen ions that lowered the pH in these locations could also be due to pedologic processes as opposed to anthropogenic sources.

Soil EC is a measure of the availability of ions in a soil sample [50] mostly associated with the concentration of dissolved ions in soil solution such as Na^{2+}, K^+, Mg^{2+}, Ca^{2+}, Cl^-, $SO_4^{2-,}$ HCO_3^- and CO_3^{2-}. Within the coffee and grass cropping systems (Fig. 2), the high EC could be attributed low use of inorganic fertilizers as coffee and grass do not require regular fertilizer application hence lower concentration of dissolved ions in soil solution. However, low EC observed in the shrubland area where there were no agricultural practices could be explained by the pedological factors. The concentration of base cations at these sites are therefore relatively higher (Fig. 2).

Soil Organic Carbon, a measure of the amount of carbon contained in the soil organic matter, is an important parameter that aids understanding of soil fertility [51]. Soil organic matter constitutes all decomposed, partly decomposed and undecomposed organic materials from plant and animal origin within the soil [12]. Decreased SOM and

Table 3 Metal pollution index (MPI) at various cropping systems (MPI < 1: less chances of pollution; 0.5 < MPI < 1: tending towards pollution; MPI > 1: surplus levels hence pollution)

Cropping system	Trace Metal element	WHO permitted soil limit ($mg.kg^{-1}$)	S_n ($mg.kg^{-1}$)	C_n ($mg.kg^{-1}$)	MPI	Pollution class interpretation
Maize	Cu	36.000	20.010	33.985	0.944	Tending towards pollution
	Pb	85.000	30.105	39.389	0.463	Less chances of pollution
	Cr (III)	100.000	8.494	11.251	0.112	Less chances of pollution
	Cd	0.800	4.535	9.042	11.303	Surplus levels hence pollution
	Zn	50.000	30.776	49.478	0.990	Tending towards pollution
	Fe	5000.000	1377.254	1583.833	0.317	Less chances of pollution
Orchard	Cu	36.000	20.010	20.152	0.560	Tending towards pollution
	Pb	85.000	30.105	29.077	0.342	Less chances of pollution
	Cr (III)	100.000	8.494	8.311	0.083	Less chances of pollution
	Cd	0.800	4.535	4.475	5.594	Surplus levels hence pollution
	Zn	50.000	30.776	29.488	0.590	Tending towards pollution
	Fe	5000.000	1377.254	1365.9	0.273	Less chances of pollution
Sugar cane	Cu	36.000	20.01	30.503	0.847	Tending towards pollution
	Pb	85.000	30.105	36.634	0.431	Less chances of pollution
	Cr (III)	100.000	8.494	11.656	0.117	Less chances of pollution
	Cd	0.800	4.535	9.347	11.684	Surplus levels hence pollution
	Zn	50.000	30.776	41.134	0.823	Tending towards pollution
	Fe	5000.000	1377.254	1474.104	0.295	Tending towards pollution
Coffee	Cu	36.000	20.01	17.855	0.496	Less chances of pollution
	Pb	85.000	30.105	31.544	0.371	Less chances of pollution
	Cr (III)	100.000	8.494	11.095	0.111	Less chances of pollution
	Cd	0.800	4.535	8.644	10.805	Surplus levels hence pollution
	Zn	50.000	30.776	43.663	0.873	Tending towards pollution
	Fe	5000.000	1377.254	1519.611	0.304	Less chances of pollution
Pastureland	Cu	36.000	20.01	14.587	0.405	Less chances of pollution
	Pb	85.000	30.105	24.532	0.289	Less chances of pollution
	Cr (III)	100.000	8.494	8.255	0.083	Less chances of pollution
	Cd	0.800	4.535	8.032	10.04	Surplus levels hence pollution
	Zn	50.000	30.776	27.543	0.551	Tending towards pollution
	Fe	5000.000	1377.254	1201.513	0.240	Less chances of pollution
Shrubland	Cu	36.000	20.01	20.010	0.556	Tending towards pollution
	Pb	85.000	30.105	30.105	0.354	Less chances of pollution
	Cr (III)	100.000	8.494	8.494	0.085	Less chances of pollution
	Cd	0.800	4.535	4.535	5.669	Surplus levels hence pollution
	Zn	50.000	30.776	30.776	0.616	Tending towards pollution
	Fe	5000.000	1377.254	1377.254	0.275	Less chances of pollution

nutrient levels have direct impacts on soil productivity, biodiversity, buffering capacity, cation exchange capacity and infiltration. Low %C recorded under maize cropping systems (Fig. 3) is due to continuous clearing of land exposes the soil to erosion that minimizes organic matter input and nutrient accumulation. In arable lands, soil organic matter content has been found to be lower compared to forested areas or natural grasslands. Routine harvesting of planted crops and their residues reduces deposition of organic materials. However, under sugar cane, coffee and grass cropping systems, the ground surfaces are covered for relative longer periods during the year which allows with continuous litter/organic matter deposition, leading to increase SOC at these sites.

Table 4 Spearman rank order correlation among the analyzed soil parameters (Fields marked* show correlation significant at $p < 0.05$)

Parameters	pH	EC	Total *N*	Na	K	Ca	Mg	Cu	Pb	Cr (III)	Cd	Zn	Fe	SOC
pH	1.00													
EC	-0.510 0.000*	1.00												
Total N	+0.03 0.967	+0.256* 0.000	1.00											
Na	-0.149 0.033	+0.204* 0.000	+0.118 0.094	1.00										
K	-0.096 0.180	+0.066 0.340	-0.161* 0.022	0.034 0.626	1.00									
Ca	-0.066 0.370	+0.059 0.403	-0.089 0.210	0.005 0.938	+0.512* 0.000	1.00								
Mg	+0.119 0.090	-0.098 0.165	-0.191* 0.010	-0.054 0.447	+0.354* 0.000	+0.446* 0.000	1.00							
Cu	-0.542** 0.000	+0.150* 0.030	-0.091 0.199	+0.013 0.849	+0.066 0.351	+0.033 0.638	-0.145* 0.039	1.00						
Pb	-0.414** 0.000	+0.326* 0.000	0.012 0.870	+0.007 0.917	+0.060 0.093	+0.141* 0.046	-0.178* 0.011	+0.528* 0.000	1.00					
Cr (III)	-0.316** 0.000	+0.280* 0.000	-0.059 0.401	-0.013 0.853	+0.232* 0.000	+0.301* 0.000	+0.157* 0.000	+0.398* 0.000	0.335** 0.000	1.00				
Cd	-0.284* 0.000	+0.200* 0.000	0.109 0.120	+0.081 0.251	+0.255* 0.000	+0.221* 0.000	+0.243* 0.000	+0.321* 0.000	0.292** 0.000	0.427** 0.000	1.00			
Zn	-0.310* 0.000	+0.347* 0.000	-0.043 0.544	+0.019 0.787	+0.375* 0.000	+0.645* 0.000	+0.234* 0.001	+0.257* 0.000	0.458 0.000	0.561** 0.000	0.345** 0.000	1.00		
Fe	-0.091 0.190	+0.159* 0.023	0.076 0.283	+0.027 0.702	+0.040 0.570	+0.153 0.030	+0.030 0.666	+0.068 0.335	0.142 0.044	0.000 0.055	0.090 0.020	0.181** 0.010	1.00	
SOC	-0.103 0.143	+0.160* 0.022	0.347** 0.000	+0.072 0.307	-0.094 0.182	-0.031 0.665	-0.040 0.569	+0.106 0.132	0.185 0.080	0.146* 0.037	0.402** 0.000	0.124 0.079	0.016 0.816	1.00

4.2 Soil trace metal elements pollution due to agriculture

Within the study area, continuous applications of the chemical pesticides and fertilizers by farmers [36] leads to the elevation of trace metal elements in the soil. This explains the observed high soil levels of trace metal elements under maize and sugarcane cropping systems (Table 3). Some pesticide and inorganic fertilizer residues are known to contain traces of TMEs such as Zinc, Copper, Nickel, Lead, Cadmium, Chromium, Arsenic and Mercury [6, 28]. Surplus residues of these agrochemicals within the soil thus raise the soil levels of TMEs. Above background concentrations, these elements pose potential health problems to plants, animals and humans, making their contamination one of the factors that restrict agricultural productivity and compromises food safety [52]. Their toxicity risks in different environmental compartments is complicated by the fact that most of them are non-biodegradable and bio-cumulative within ecosystems, thus persisting in biological systems and have the potential of being transferred along trophic levels can be toxic to plants, animals and humans. This finding concurs with [53] which reported that in Wanshan Mining District of China, soils under corn field production were slightly contaminated with lead (22–100 mg/kg), copper (31–64 mg/kg), and nickel (22–76 mg/kg) and moderately contaminated with zinc (112–635 mg/kg). Maize production cropping systems are hence more prone to soil trace metal element contamination (Table 3).

Contrary to what was expected, even non-cultivated sites in the study area also recorded trace metal elements residues (Table 3). Relatively high concentrations of TMEs in control sites is attributable to pedologic processes. Under normal circumstances, these elements occur naturally in the environment but at very low concentrations (background concentration). However, their natural release rate within the soil is influenced by a combination of factors such as chemical composition of the parent rock, the prevailing climatic and biotic factors that mediate pedologic and geologic process through which associated with trace metal element exudation [54]. Despite the various natural sources of trace metal elements in the environment, many published works have documented other anthropogenic processes and activities that may also increase trace metal elements concentrations within the soil such as industrialization, agriculture, mining, waste discharge and combustion of fossil fuels [27, 55–56]. Due to the insignificant levels of mining and industrialization in the study area, surplus levels of trace metal element, above concentrations, in the study area can be attributed to agricultural activities.

In many parts of the world, many empirical studies have documented trace metal element influxes in agricultural soils due to application of various classes of agrochemicals such as fertilizers and pesticides [7, 57–58]. High application of inorganic fertilizers such as lime, phosphatic, nitrate and potash fertilizers has been found to contribute to elevated levels of Chromium, Cadmium, Manganese, Zinc, Copper, Lead and Nickel [7]. This explains the high mean levels of Mercury, Copper and Zinc recorded under maize and sugarcane cropping systems in the study area (Table 4). With the introduction of fertilizer subsidy under Agricultural Sector Support Programme and other county government initiatives in Kenya, most farmers in the study area have embraced the use of inorganic fertilizers due to the perceived conferment of high yields [36]. These inorganic fertilizers re applied by farmers during planting and later for top dressing to boost productivity. Previously [46], reported that top soil samples collected from Nzoia sugarcane nucleus showed high levels of trace metal elements in the order of 42.38 mg/Kg,

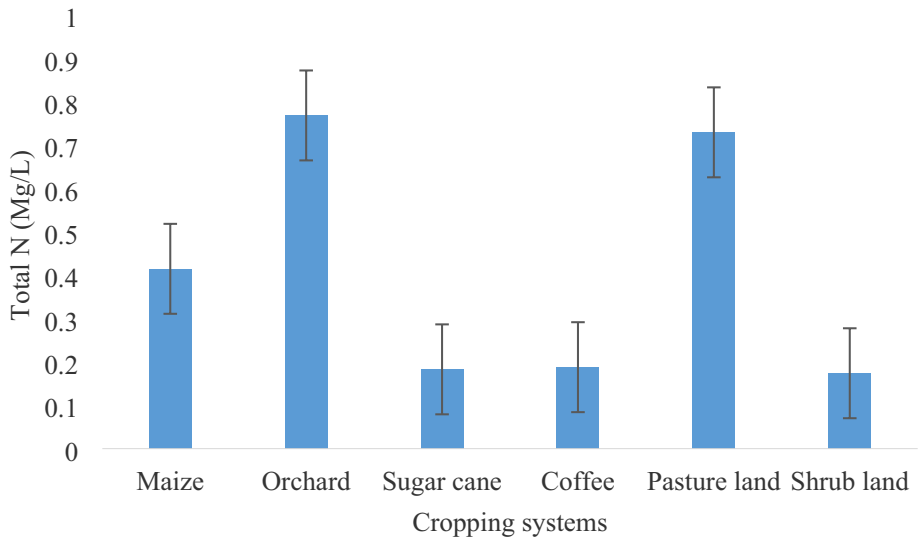

Fig. 5 Variation in total N across cropping systems

59.12 mg/Kg, 116.27 mg/Kg and 409.84 mg/Kg of Cr, Pb, Cu, Zn and Fe, respectively. The study concluded that within the sugarcane farms, the trace metal element concentrations were above the international standards. Moreover [59], after investigation trace metal elements in vineyard and orchard soils, concluded that besides the application of inorganic fertilizers, the use of foliar fungicides in orchards can elevate the concentrations of trace metals such as copper and zinc upto toxicity levels for fruit trees and cover crops. More recently [55], reported that mean concentrations of Pb, Cr, Hg, As, and Se in soils sampled from cultivated areas exceeded the background values of Guizhou Province in China by 1.12, 1.28, 2.36, 1.27, and 2.4 times, respectively.

As fertilizer application is intensified, accumulation of trace metal element residues in the soil rises. This is why a high MPI value of 11.303 was observed within the maize cropping system with respect to cadmium (Table 3). Cadmium is one of the trace metal elements, traces of which have been associated with chemical pesticide and fertilizer residues. This finding concurs with [60] that found out that even though soils sampled from orchards did not have trace metal element pollution threats with respect to Cu, Cr, Zn, Hg and Zn, 10.0% of the soil samples collected from the orchards were under threat from cadmium pollution with MPI ≥ 1. Even though cadmium is one of the nonessential plant elements, it is quite ubiquitous in the environment with its sources associated with human activities such as use of phosphate fertilizers. However, soil contamination through cadmium saturation may have adverse effects on plant physiology and possible transference to other trophic levels [61]. Increased cadmium concentrations inhibits absorption and translocation of essential plant nutrients and water while also exposes plants to oxidative damage.

In orchards, there is little routine application of inorganic fertilizers during the growing seasons compared to maize and sugarcane. This explains why soils sampled from the orchards recorded showed trace metal elements concentrations that that are intermediate between maize cropping system and the control sites. In china, empirical analysis found that even though soil trace metal element pollution in agricultural soils irrigated with sewage in Wuqing, Tianjin was mild except for Cd and Pb, PCA and PMF showed that soil parental material contributed significantly to Ca, Mg (23.46%) while agricultural

activities contributed (29.97%) of Pb, Cu, and Zn pollution [62]. In the Chinese Shaanxi region [60], observed that concentration of Cr, Cu, Cd, Hg and Pb in orchard soils did not pose any soil pollution threats with MPI ≤ 1. However, 10.0% of the soil samples were under cadmium pollution. Besides the use of inorganic fertilizers, different types of pesticides including herbicides, fungicides and insecticides have also been found to contain traces of trace metal elements such as Zinc, Copper, Nickel, Lead, Cadmium, Chromium, Arsenic and Mercury [57]. With repeated application of these pesticides in large scale farms in the study area, there is continued deposition of trace metal element traces in agricultural soils posing potential ecological risks.

Combined trace metal element residues from inorganic fertilizers and pesticides in these large scale farms therefore leads to elevated levels of the analyzed elements in the soil (Table 2). This is why higher levels of trace metal elements were recorded under cultivated areas compared to the grassland and control plots. In the Wanshan Mining District of China [53], reported that soil under corn cropping system were moderately contaminated with lead (22–100 mg/kg), copper (31–64 mg/kg), and nickel (22–76 mg/kg) and moderately contaminated with zinc (112–635 mg/kg). Although the use of organic manures has been advocated for by environmentalists due to the perceived minimal environmental impacts, previous works have shown that application of biosolids and manures can lead to accumulation of Zinc, Nickel, Copper, Lead, Cadmium, Chromium and Mercury in agricultural soils [7, 32].

5 Conclusions

In conclusion, there were spatial variations in trace metal element concentrations across cropping systems. In comparison with the established international standards, the concentrations of the analysed trace metal elements except Cadmium in all cropping systems considered were lower than the globally permitted levels in agricultural soils. Based on calculated MPI indices, the rank order pollution potential associated with trace metal contamination in all cropping system was Cd > Zn > Cu > Pb > Fe > Cr. Therefore, Zn, Cu, Pb, Fe and Cr did not pose any soil pollution threat.

Under maize, sugarcane, coffee and pastureland cropping systems, the concentrations of Cd were over ten times higher than the WHO permitted levels (0.8 $mg.kg^{-1}$) in agricultural soils. There is therefore need to regulate application of Cd-containing agrochemicals and the types of crops planted in successive rotations to minimize chances of Cadmium transfer to other trophic levels.

However, in orchards and shrublands, Cd concentrations were only about six times higher than the WHO permitted levels (0.8 $mg.kg^{-1}$) in agricultural soils. This necessitates the need for continuous monitoring of soil Cadmium levels to minimize further soil Cd-enrichment.

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Author contributions

All the authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by Zedekiah Odira (First Author). The first draft of the manuscript was written by Zedekiah Odira and Elizabeth Omukunda and proofread by Solomon Omwoma and Peter Otieno, and it was reviewed by Odipo Osano and Sandra Khatiebi, and overall supervised by Joseph Lalah. All authors read and approved the final manuscript.

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Data availability
The authors declare that the data supporting the findings of this study are available within the paper. The study did not use secondary data available from other platforms. Site-specific soil trace metal element concentration and physicochemical data collected during the study are available with the corresponding author who upon reasonable request, can make them available.

Declarations

Ethics approval and consent to participate
The study was approved by the institutional Ethics Review Board of Masinde Muliro University of Science and Technology, approval number MMUST/ISERC/122/2025.

Consent for publication
All authors have read and approved the final version of this manuscript and consent to its publication in the Journal of Discover Soil.

Competing interests
The authors declare no competing interests.

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